SALA HACKATHON 2026

RainCaster Galapagos Dataset

Data Description & Hackathon Guide

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1. Overview

Welcome to the Galapagos RainCaster Challenge.

On San Cristóbal Island, the Institute of Geography at Universidad San Francisco de Quito (USFQ) in collaboration with the the Galápagos Science Center (GSC), operates a network of four meteorological stations: El Mirador, Cerro Alto, El Junco, and Merceditas. Recording since June 2015, this network continuously transmits live data every 15 minutes. You can explore the live data visualization platform here: https://weathergsc.usfq.edu.ec/. The locations of these stations are shown in Figure 1.

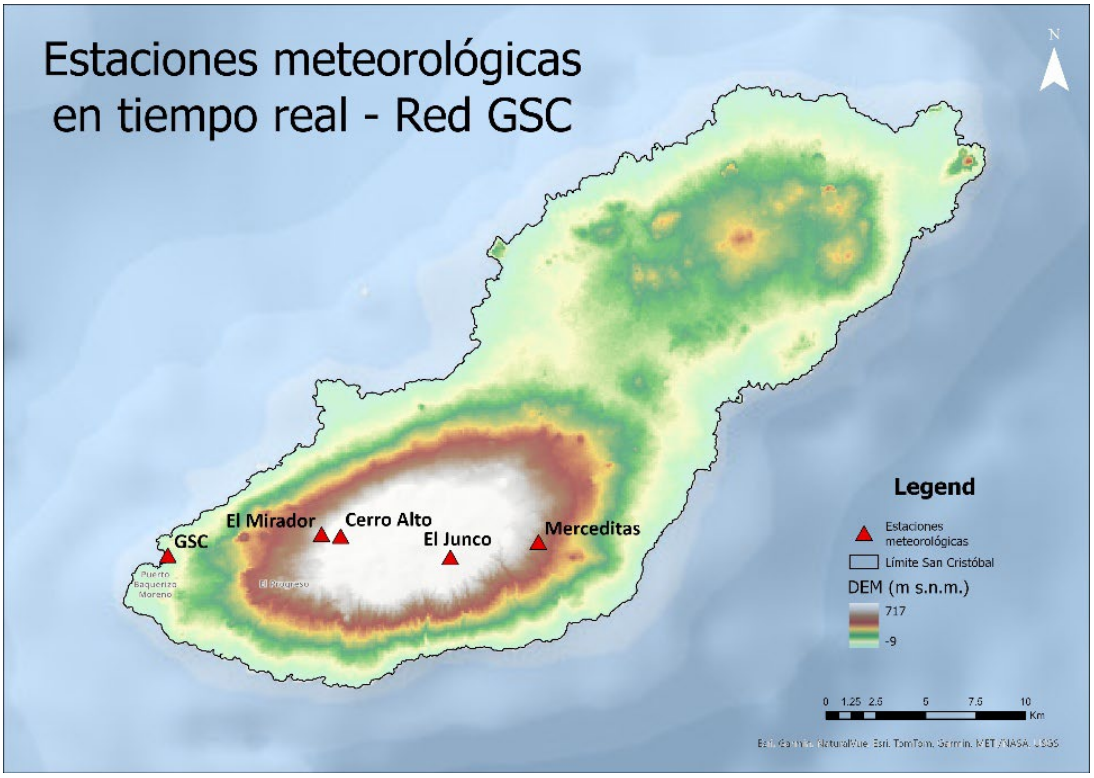

Figure 1. Map of weather stations in San Cristóbal, Galapagos

Your task is to leverage this high-resolution, real-time data stream to develop a robust model capable of predicting precipitation events at each weather station. The goal is to build a system that acts as a live early warning application for the island.

To ensure a rigorous evaluation process, all participating teams must adhere to the following guidelines and submission formats.

2. Data Description

2.1 Dataset Summary

	Details
Time period	*June 2015 – March 2026*
Temporal resolution	*15-minute intervals*
Spatial coverage	*4 weather stations across San Cristóbal Island, Galapagos*
Number of records	*~1,500,000 total (~375,000 per station)*
File format(s)	*CSV*
Total data volume	*~300 MB*
Update frequency	*Ongoing (real-time updates every 15 minutes)*

2.2 Variables

Variable	Description	Unit	Record Start	Stations
TIMESTAMP	Date and time of registration (15-minutes intervals). Time zone: UTC-6	month/day/year hour:minute	June -2015	Mirador, Cerro Alto, Junco, Merceditas
Rain_mm_Tot	Total precipitation. Cumulative rainfall over the measurement period. 1 mm = 1 litre/m^2	mm	June -2015	Mirador, Cerro Alto, Junco, Merceditas
AirTC_Avg	Average air temperature	° C	June -2015	Mirador, Cerro Alto, Junco, Merceditas.
LWmV_Avg	Longwave radiation - average of raw signal: Mean incoming infrared radiation from the atmosphere	mV	June -2015	Mirador, Cerro Alto, Junco, Merceditas
RH_Max	Maximum relative humidity: Peak percentage of water vapor in the air	%	June -2015	Junco, Merceditas.
SlrkW_Avg	Average global solar radiation: Mean total incoming shortwave solar radiation	kW	June -2015	Mirador, Cerro Alto, Junco, Merceditas
SlrMJ_Tot	Total solar radiation: Cumulative incoming solar energy over the period	MJ	June -2015	Mirador, Cerro Alto, Junco, Merceditas
NR_Wm2_Avg	Average net radiation: Mean balance between incoming and outgoing radiation	W/m^2	June -2015	Mirador, Cerro Alto, Junco, Merceditas
VW /VW_Avg	Volumetric water content in the first soil horizon: Water volume per soil volume in topsoil layer	Dimensionless value between 0 and 1	June -2015	Mirador, Cerro Alto, Junco, Merceditas
PA_uS /PA_uS_Avg	Electrical conductivity in the first soil horizon: Soil salinity or nutrient level in topsoil layer	µS	June -2015	Mirador, Cerro Alto, Junco, Merceditas
VW_2 / VW_2_Avg	Volumetric water content in the second soil horizon: Water volume per soil volume in second soil layer	Dimensionless value between 0 and 1	June -2015	Mirador, Cerro Alto, Junco, Merceditas
PA_uS_2/ PA_uS_2_Avg	Electrical conductivity in the second soil horizon: Soil salinity or nutrient level in second soil layer	µS	June -2015	Mirador, Cerro Alto, Junco, Merceditas

VW_3 / VW_3_Avg	Volumetric water content in the third soil horizon: Water volume per soil volume in third soil layer	Dimensionless value between 0 and 1	June -2015	Mirador, Cerro Alto, Junco, Merceditas
PA_uS_3 / PA_uS_3_Avg	Electrical conductivity in the third soil horizon: Soil salinity or nutrient level in third soil layer	µS	June -2015	Mirador, Cerro Alto, Junco, Merceditas
WS_ms_Avg	Average wind speed: Mean air velocity	m/s	June -2015	Mirador, Cerro Alto, Junco, Merceditas
WindDir	Wind direction: Predominant direction wind originates from	Degrees [°] between 0 and 360	June -2015	Cerro Alto, Merceditas.
AirTC_Max	Maximum air temperature: Highest recorded ambient air temperature	° C	March - 2025	Mirador, Junco, Merceditas
AirTC_Min	Minimum air temperature: Lowest recorded ambient air temperature	° C	March - 2025	Mirador, Junco, Merceditas
RH_Min	Minimum relative humidity: Lowest percentage of water vapor in the air	%	March - 2025	Junco, Merceditas.
RH_Avg	Average relative humidity: Mean percentage of water vapor in the air	%	March - 2025	Junco, Merceditas.
SlrW_Max	Maximum solar radiation: Peak incoming shortwave solar radiation	W	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
SlrW_Min	Minimum solar radiation: Lowest incoming shortwave solar radiation	W	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
LWMDry_Tot	Leaf wetness minutes – dry: Duration leaf surface is dry	minute	March - 2025	Mirador, Junco.
LWMCon_Tot	Leaf wetness minutes– condensation: Duration time moisture condenses on leaf surface	minute	March - 2025	Mirador, Junco.
LWMWet_Tot	Leaf wetness minutes – wet: Duration leaf surface is wet from rain or dew	minute	March - 2025	Mirador, Junco.
LWmV_Max	Maximum longwave radiation: Highest incoming infrared radiation from the atmosphere	mV	March - 2025	-
LWmV_Min	Minimum longwave radiation: Lowest incoming infrared radiation from the atmosphere	mV	March - 2025	-
WS_ms_Max	Maximum wind speed: Highest recorded horizontal air velocity	m/s	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
WS_ms_Min	Minimum wind speed: Lowest recorded horizontal air velocity	m/s	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
WindDir_Max	Maximum wind direction: Direction associated with peak wind speed	Degrees [°] between 0 and 360	March - 2025	Cerro Alto, Merceditas.
WindDir_Min	Minimum wind direction: Direction associated with lowest wind speed	Degrees [°] between 0 and 360	March - 2025	Cerro Alto, Merceditas.
WindDir_Avg	Average wind direction: Mean wind origin direction	Degrees [°] between 0 and 360	March - 2025	Cerro Alto, Merceditas.
NR_Wm2_Max	Maximum net radiation: Peak balance between incoming and outgoing radiation	W/m^2	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
NR_Wm2_Min	Minimum net radiation: Lowest balance between incoming and outgoing radiation	W/m^2	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
CNR_Wm2_Max	Maximum corrected net radiation: Peak adjusted balance of incoming and outgoing radiation	W/m^2	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
CNR_Wm2_Min	Minimum corrected net radiation: Lowest adjusted balance of incoming and outgoing radiation	W/m^2	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
CNR_Wm2_Avg	Average corrected net radiation: Mean adjusted balance of incoming and outgoing radiation	W/m^2	March - 2025	Mirador, Cerro Alto, Junco, Merceditas
RECORD	Registration number: Unique identifier for the data record or station	-	June -2015	Mirador, Cerro Alto, Junco, Merceditas
BattV_Avg	Average battery voltage: Mean voltage level of the power supply system	Volts	June -2015	Mirador, Cerro Alto, Junco, Merceditas
BattV_Min	Minimum battery voltage: Lowest recorded voltage level of the power supply system	Volts	March - 2025	Mirador, Cerro Alto, Junco, Merceditas

2.3 File Structure & Access

The dataset is organized into four separate comma-separated values (.csv) files, with each file containing the continuous 15-minute time-series data for a single meteorological station on San Cristóbal Island.

File Naming Conventions: Each file is named using an abbreviated station code followed by "_consolid_f15.csv" (indicating consolidated 15-minute frequency data). The four files you will download are:

- CER_consolid_f15.csv
- JUN_consolid_f15.csv
- MERC_consolid_f15.csv
- MIRA_consolid_f15.csv

Mandatory Station ID Codes: For all modeling tasks, data processing, and final submissions during the hackathon, teams must use the specific id codes assigned to each station. The mapping is as follows:

- *cer* corresponds to the Cerro Alto station
- *jun* corresponds to the El Junco station
- *merc* corresponds to the Merceditas station
- *mira* corresponds to the El Mirador station

Station Metadata Table: To assist with any spatial analysis or feature engineering, the following table provides the exact geographical coordinates (using the WGS 1984 geographic coordinate system) and altitude for each of the four stations using their official ID codes.

Name	id	filename	Longitude	Latitude	Altitude (m a.s.l.)
Merceditas	merc	MERC_consolid_f15.csv	-89.44202039	-0.889712315	100
El Junco	jun	JUN_consolid_f15.csv	-89.48162446	-0.896537076	548
Cerro Alto	cer	CER_consolid_f15.csv	-89.53098555	-0.887048868	517
El Mirador	mira	MIRA_consolid_f15.csv	-89.53958685	-0.886247558	387

3. General Rules & Modeling Constraints

- Data Usage: Models must be primarily trained on the provided historical dataset. If external datasets (e.g., satellite imagery, global climate indices) are used, they must be publicly available, open-source, and explicitly documented in your final presentation.
- Look-ahead Bias: Your model must strictly predict future states (+1 hour, +3 hours, +6 hours) using only data available up to the current timestamp (T_0). To keep things fair and accurate, please be careful to avoid data leakage (when future information, or data not available at the time of prediction, is included in the training set). Submissions containing leakage unfortunately can't be scored, so double-check your process to stay in the running!
- Handling Missing Data: The sensor network occasionally experiences downtime or missing values. Your pipeline must include a robust imputation or handling strategy, so your evaluation loop does not crash when encountering NaN values.

4. The evaluation Strategy: Walk-Forward Validation

Time-series models cannot be evaluated using random data splits. To mimic real-world deployment, teams must implement a chronological split using *Rolling Walk-Forward Validation* over the final year of the dataset.

- The Split: You must isolate the final 365 days of the provided dataset. This final year is your evaluation period (Validation/Test). All data prior to this 1-year mark is your initial Training set.
- The Walk-Forward Loop: While the dataset has a 15-minute resolution, forecasts will be evaluated on an hourly basis. Starting at the very first top-of-the-hour timestamp (e.g., 00:00:00) of that final year, your model must:

  1. Ingest all available historical 15-minute data up to that exact moment (T_0).
  2. Generate predictions (classes and probabilities) for the +1h, +3h, and +6h windows.
  3. Step forward by 1 hour (e.g., from 08:00:00 to 09:00:00) to the next forecasting timestamp.
  4. The true 15-minute data points from the previous hour are now "revealed" and added to your available historical context for the next prediction.

5. Evaluation metrics & Scoring

Predicting precipitation presents a severe "class imbalance" problem. It does not rain most of the time. Therefore, traditional accuracy is not a valid metric; a model could achieve 90% accuracy simply by predicting "No Rain" every hour. To evaluate the true predictive power of your models, teams are required to calculate performance metrics across the final 365-day walk-forward period by comparing their predictions against the ground-truth values of that final year.

Task: Classification

Target Variable: A multi-class label for the expected accumulated precipitation within the time window. The total accumulated precipitation (in mm) within each forecasting time horizon must be categorized into one of three distinct classes: 0 (No rain), 1 (Light rain), or 2 (Heavy rain). The specific thresholds for these classes vary depending on the prediction window, as detailed in the table below. Note that the limits defining "Light rain" and "Heavy rain" scale proportionally with the length of the time horizon.

Time window	Class	Sum of precipitation in time horizon (mm)
+1 hour	0: No rain	0
	1: Light rain	0 - 0.254
	2: Heavy rain	> 0.254
+3 hours	0: No rain	0
	1: Light rain	0 - 0.508
	2: Heavy rain	> 0.508
+6 hours	0: No rain	0
	1: Light rain	0 - 0.762
	2: Heavy rain	> 0.762

Required Evaluation Metrics Table:

For each weather station and time window, you must calculate these metrics: Macro F1, Micro F1, Weighted F1, precision, recall, f1-score, support, accuracy, macro avg, weighted avg. Include the results in your final ReadMe documentation. To calculate this metrics, you can use *"from sklearn.metrics import f1_score, classification_report"*. The following is an example:

Macro F1:	0.65
Micro F1:	0.67
Weighted F1:	0.64

	precision	recall	f1-score	support
0	0.67	1	0.8	2
1	0.5	0.33	0.4	3
2	0.75	0.75	0.75	4
accuracy			0.67	9
macro avg	0.64	0.69	0.65	9
weighted avg	0.65	0.67	0.64	9

Optional additional metrics: While the table above contains the mandatory metrics for judging, teams are highly encouraged to calculate and report any other evaluation metrics that they consider relevant to further demonstrate the robustness and operational value of their models.

6. Required Submission Format

Your final output must consist of three separate comma-separated values (.csv) files, one for each forecasting horizon: submission_1h.csv, submission_3h.csv, and submission_6h.csv.

Every file must include both your model's predictions and the actual observed ground-truth values for that specific time window.

Required Columns for all three files:

- timestamp: The current time in standard ISO 8601 format (e.g., YYYY-MM-DD HH:MM:SS). Must be exactly on the hour (e.g., 14:00:00).
- station_id: The identifier for the station.
- pred_class: Multi-class prediction (0, 1, or 2).
- pred_prob: The model's probability for the chosen intensity class (e.g., 0.65 for 65%).
- obs_class: The actual intensity class ground truth for the window (0, 1,or 2).
- obs_precip_mm: The actual total accumulated precipitation (in mm) recorded during the forecast window.

Example of the structure of *submission_3h.csv*:

timestamp	station_id	pred_class	pred_prob	obs_class	obs_precip_mm
2026-02-17 14:00:00	station_1	2	0.65	2	4.318
2026-02-17 15:00:00	station_1	0	0.84	0	0
2026-02-17 16:00:00	station_1	1	0.56	0	0
2026-02-17 17:00:00	station_1	0	0.61	0	0
2026-02-17 14:00:00	station_3	1	0.67	0	0
2026-02-17 15:00:00	station_3	0	0.78	0	0
2026-02-17 16:00:00	station_3	0	0.54	1	0.508
2026-02-17 17:00:00	station_3	0	0.71	0	0

In this example, for the +3h window made at 14:00:00 for station 1, the model predicted Class 2 with 65% probability, and the actual observed data was 4.318 mm, confirming it was indeed a Class 2 event.

7. Deliverables for final judging

By the hackathon deadline, your team repository must include:

1. The Completed Evaluation Metrics Table: Clearly displayed in your ReadMe.
2. The submission.csv files: Containing the predictions for the final year.
3. requirements.txt or environment.yml: Listing all dependencies to run your code.
4. pipeline.py or Jupyter Notebook: The fully reproducible code demonstrating your train/validation split, your walk-forward evaluation loop, and the calculation of your final metrics.
5. ReadMe Documentation: A brief overview of your chosen architecture, your approach to class imbalance, and instructions on how to run your code.

8. Real-World impact

A prototype from this challenge will deliver value by transforming meteorological data into a proactive Early Warning System for San Cristobal, Galapagos. As a globally vital and highly sensitive ecosystem, the Galapagos faces vulnerabilities to sudden, extreme weather shifts. An accurate precipitation forecasting tool would serve as a critical asset across multiple sectors: in tourism, operators could use short-term forecasts to adjust boat schedules or hiking routes to ensure visitor safety; in agriculture, local farmers in the highlands could make rapid, data-driven decisions regarding crop protection and

water management. Because the GSC-USFQ sensor network is already actively transmitting live data, the path from hackathon prototype to deployment is straightforward. A winning model could be directly integrated into the existing data pipeline, serving as the backend for the dashboard that we manage (https://weathergsc.usfq.edu.ec/) that empowers the local community to anticipate risks rather than merely react to them.

Furthermore, this challenge presents an opportunity for academic contribution. The methodologies developed by the winning team could serve as a framework for a peer-reviewed scientific publication. The winning participants would be invited to collaborate as co-authors, contributing their innovations to the scientific literature on precipitation short-term forecasting.

9. Resources & References

TensorFlow Time Series Forecasting Tutorial: This guide covers essential techniques for forecasting continuous data using deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. It is highly relevant for this challenge, as it provides practical code examples on how to handle multivariate time-series datasets, build data windows, and predict future time steps.

Link: https://www.tensorflow.org/tutorials/structured_data/time_series]